Chapter 13, Problem 24P

Problem

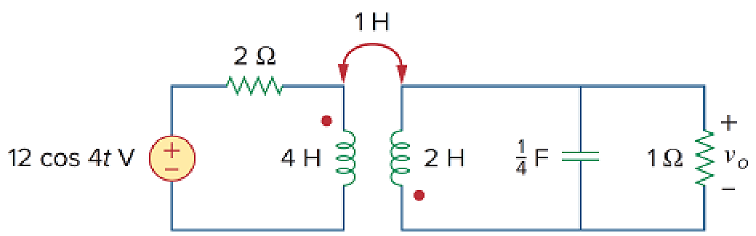
In the circuit of Fig. 13.93,

(a) find the coupling coefficient,

(b) calculate *vo*,

(c) determine the energy stored in the coupled inductors at *t* = 2 s.

Figure 13.93



Step-by-step solution

Step 1 of 11

(a)

Refer to circuit diagram in Figure 13.93 in the text book.

Determine the value of coupling coefficient.

$$k = \frac{M}{\sqrt{L_1 L_2}}$$

Substitute 4 H for L_1, 2 H for L_2 and 1 H for *M*.

$$\begin{aligned} k &= \frac{1}{\sqrt{(4)(2)}} \\ &= \frac{1}{\sqrt{8}} \\ &= 0.3535 \end{aligned}$$

Therefore, the value of coupling coefficient k is $\boxed{0.3535}$.

Step 2 of 11

(b)

The value of frequency is,

$$\omega = 4\,\text{rad/s}$$

Calculate the impedance value of the 4 H inductor.

$$\begin{aligned} j\omega L_1 &= j(4)(4) \\ &= j16\ \Omega \end{aligned}$$

Calculate the impedance value of the 2 H inductor.

$$\begin{aligned} j\omega L_2 &= j(4)(2) \\ &= j8\ \Omega \end{aligned}$$

Calculate the impedance value of the 1 H mutual inductor.

$$\begin{aligned} j\omega M &= j(4)(1) \\ &= j4\ \Omega \end{aligned}$$

Calculate the impedance value of the capacitor.

$$\begin{aligned} \frac{1}{j\omega C} &= \frac{1}{j(4)\left(\frac{1}{4}\right)} \\ &= -j\ \Omega \end{aligned}$$

Step 3 of 11

The frequency domain circuit is shown in Figure 1.

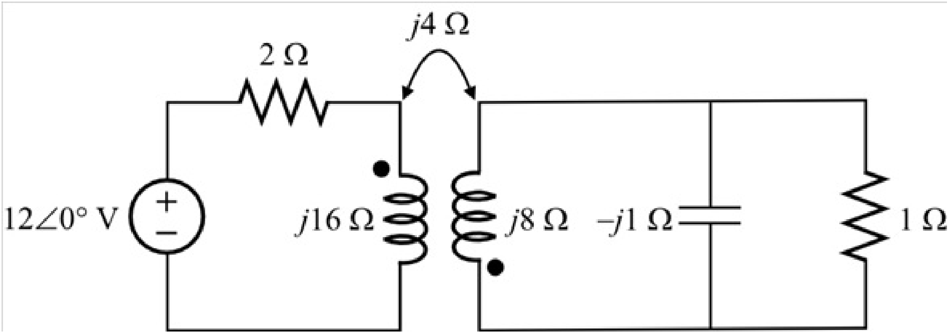


Figure 1

Step 4 of 11

In the circuit, capacitor and $1\ \Omega$ resistor are connected in parallel. Calculate the equivalent impedance.

$$\begin{aligned} Z_{eq} &= (1\ \Omega) \| (-j\ \Omega) \\ &= \frac{(1\ \Omega)(-j\ \Omega)}{1 - j\ \Omega} \\ &= \frac{-j\ \Omega}{1-j} \\ &= 0.5(1-j)\ \Omega \end{aligned}$$

Step 5 of 11

The reduced frequency domain circuit is shown in Figure 2.

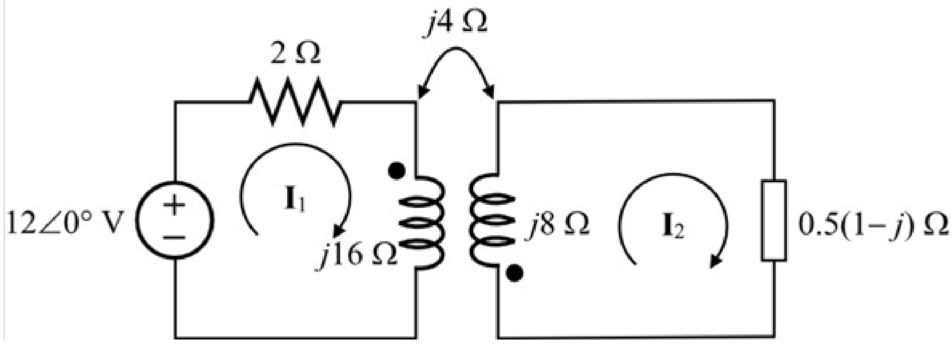


Figure 2

Step 6 of 11

Apply Kirchhoff's voltage law to first loop.

$$-12\angle 0° + 2(\mathbf{I}_1) + (j16)(\mathbf{I}_1) + (j4)(\mathbf{I}_2) = 0$$
$$(2 + j16)(\mathbf{I}_1) + (j4)(\mathbf{I}_2) = 12$$
$$(1 + j8)\mathbf{I}_1 + (j2)\mathbf{I}_2 = 6 \quad \text{...... (1)}$$

Apply Kirchhoff's voltage law to second loop.

$$(j4)\mathbf{I}_1 + (0.5 - j0.5)\mathbf{I}_2 + (j8)\mathbf{I}_2 = 0$$
$$(j4)\mathbf{I}_1 + (0.5 + j7.5)\mathbf{I}_2 = 0$$
$$(j4)\mathbf{I}_1 = -(0.5 + j7.5)\mathbf{I}_2$$
$$\mathbf{I}_1 = -\left(\frac{0.5 + j7.5}{j4}\right)\mathbf{I}_2 \quad \text{...... (2)}$$

Step 7 of 11

Recall equation (1).

$$(1 + j8)\mathbf{I}_1 + (j2)\mathbf{I}_2 = 6$$

Substitute $-\left(\frac{0.5 + j7.5}{j4}\right)\mathbf{I}_2$ for $\mathbf{I}_1$.

$$-(1 + j8)\left(\frac{0.5 + j7.5}{j4}\right)\mathbf{I}_2 + (j2)\mathbf{I}_2 = 6$$
$$-(2.875 + j14.875)\mathbf{I}_2 + (j2)\mathbf{I}_2 = 6$$
$$-(2.875 + j12.875)\mathbf{I}_2 = 6$$
$$\mathbf{I}_2 = \frac{-6}{2.875 + j12.875}$$
$$= -0.099 + j0.4438 \text{ A}$$

Step 8 of 11

Determine the value of output voltage.

$$\mathbf{V}_o = 0.5(1-j1)\mathbf{I}_2$$

Substitute $-0.099+j0.4438\text{ A}$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{V}_o &= 0.5(1-j1)(-0.099+j0.4438)\\ &= 0.1724+j0.2715\\ &= 321.6\angle -57.58^\circ\text{ mV}\end{aligned}$$

The output voltage in time domain format is,

$$v_o(t) = 321.6\cos(4t-57.58^\circ)\text{ mV}$$

Therefore, the output voltage $v_o(t)$ is $\boxed{321.6\cos(4t-57.58^\circ)\text{ mV}}$.

Step 9 of 11

(c)

Determine the value of current $\mathbf{I}_1$.

$$\mathbf{I}_1 = -\left(\frac{0.5+j7.5}{j4}\right)\mathbf{I}_2$$

Substitute $-0.099+j0.4438\text{ A}$ for $\mathbf{I}_2$.

$$\begin{aligned}\mathbf{I}_1 &= -\left(\frac{0.5+j7.5}{j4}\right)(-0.099+j0.4438)\\ &= -0.13+j0.8445\\ &= 0.8545\angle -81.21^\circ\text{ A}\end{aligned}$$

$$i_1(t) = 0.8545\cos(4t-81.21^\circ)\text{ A}$$

Calculate the current $i_1(t)$ at $t=2\text{ s}$.

$$\begin{aligned}i_1(t) &= 0.8545\cos\left((4)(2)-(81.21^\circ)\left(\frac{\pi}{180^\circ}\right)\right)\text{ A}\\ &= 0.8545\cos(8-1.417)\text{ A}\\ &= (0.8545)(0.9556)\text{ A}\\ &= 0.8165\text{ A}\end{aligned}$$

Step 10 of 11

The current $\mathbf{I}_2$ is,

$$\begin{aligned}\mathbf{I}_2 &= -0.099+j0.4438\text{ A}\\ &= 0.4547\cos(4t+102.59^\circ)\text{ A}\end{aligned}$$

$$i_1(t) = 0.4547\cos(4t+102.59^\circ)\text{ A}$$

Calculate the current $i_2(t)$ at $t=2\text{ s}$.

$$\begin{aligned}i_2(t) &= 0.4547\cos\left((4)(2)+(102.59^\circ)\left(\frac{\pi}{180^\circ}\right)\right)\text{ A}\\ &= 0.4547\cos(8+1.79)\text{ A}\\ &= (0.4547)(-0.9338)\text{ A}\\ &= -0.4246\text{ A}\end{aligned}$$

Step 11 of 11

Determine the energy stored in the coupled inductors at $t = 2\text{ s}$.

$$\begin{aligned} W &= \frac{1}{2}L_1 i_1^2 + \frac{1}{2}L_2 i_2^2 + M i_1 i_2 \\ &= \frac{1}{2}(4)(0.8165)^2 + \frac{1}{2}(2)(-0.4246)^2 + (1)(0.8165)(-0.4246) \\ &= 1.333 + 0.1803 - 0.3467 \\ &= 1.167\text{ J} \end{aligned}$$

Therefore, the energy stored in the coupled inductors at $t = 2\text{ s}$ is $\boxed{1.167\text{ J}}$.